

The code for \$a_2\$ is:

Please provide a brief explanation of your answer.

Evaluating the sum in $f(n)$ gives

$$\lim_{n \rightarrow \infty} f(n) = \frac{1}{4} + 72^{-10^{24}} - \sum_{j=3}^{\infty} 72^{-10^{23} \cdot j}.$$

Let $y = 72^{-10^{23}}$; then the constant term is y^{10} and the subtracted sum is $\frac{y^3}{1-y}$. Since $0 < y \ll 1$, $y^{10} \ll y^3$, making the overall addition negative. Hence $a_2 = 1 + 2\left(y^{10} - \frac{y^3}{1-y}\right) < 1$ (incredibly close to 1, but strictly less).

The optimization problem is to minimize $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ and $x_1, x_2 \geq 0$. With $a_2 < 1$, the unique minimizer is $(x_1^*, x_2^*) = (1, 0)$.

The infinity-norm distance from Jordan's point $(0.3, 0.7)$ to $(1, 0)$ is $\max(|0.3 - 1|, |0.7 - 0|) = 0.7$, which is **not** ≤ 0.4 . (The revealed statement about $(0.6, 0.4)$ is true: its distance to $(1, 0)$ is $0.4 \leq 0.45$.)